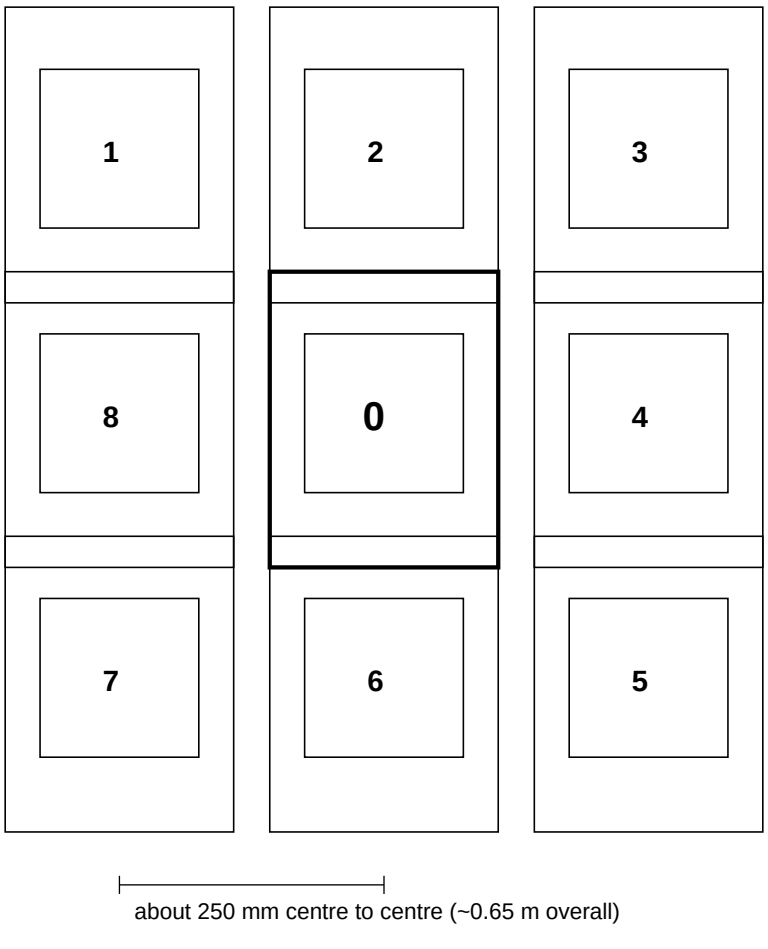
Floor reference: how to lay it out

1. Print this PDF on US Letter at 100 % (Actual size, NOT 'fit to page'); keep this page for yourself.
2. Measure the black square of sheet 0 (should be 150 mm) and enter it, in metres, as marker.edge_m.
3. Put sheet 0 in the middle, directly under where the claw rests at startup.
4. Lay the other 8 sheets as drawn below, 250 mm centre to centre (tag centre to tag centre; never more than 280 mm: the resting claw must see at least TWO tags). Exact spacing and rotation do not matter: the robot SURVEYS the layout itself during startup (step 7). From the resting claw it surveys the sheets its camera sees (sheet 0 and usually 2-3 neighbours, ~30 frames); the others are surveyed while the claw hangs and calibrates, as they come into view. A sheet that is never seen clearly is NOT surveyed and NOT used. A closer, tighter grid is better than a wide one. Letter sheets are 279 mm tall, so at this pitch sheets above/below each other OVERLAP by about 29 mm (side by side they leave a 34 mm gap). Overlap only the blank paper margins (the header with the big number at the top, the scale bar and text at the bottom); NEVER cover any part of a tag or its white border.
5. Every sheet must lie FLAT and fully visible: tape all corners down, no curl, no folds, no objects or cables across a tag, avoid glare (close curtains / matte paper). Carpet ridges and rugs that slide are not OK.
6. Sheet 0 defines the frame: its +X points right as printed. Enter the approximate room heading of that +X as marker.heading_prior_rad.
7. Layout file: set marker.board_layout_file to a layout file that exists (make the starting one, only sheet 0 known, with `python -m tools.print_floor_board --layout-json data/floor_board.json --no-pdf`). The robot rewrites it after surveying and keeps the previous contents as <file>.bak. Set marker.edge_m to the measured edge of sheet 0 (step 2); the layout must agree.
8. Do NOT move or bump any sheet during a run: nothing re-surveys it until the next startup. A sheet that is in view at the next startup and has moved by more than a few mm is found and surveyed again; a sheet out of view is trusted as saved.

Schematic, looking down at the floor (not to scale; the 0 sheet is the thick one)

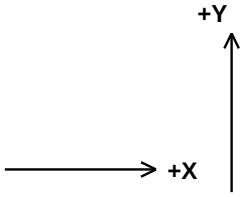


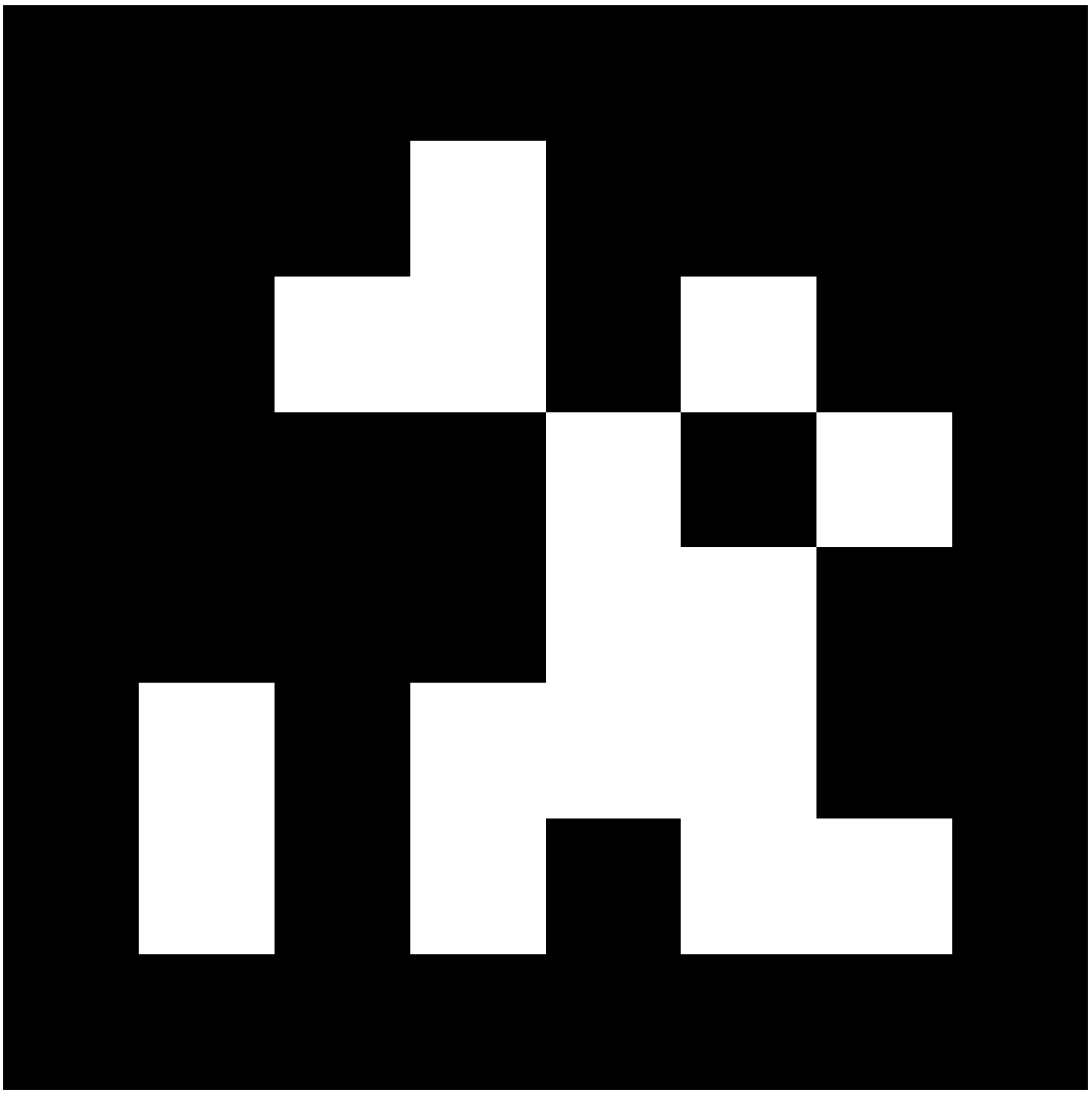
0

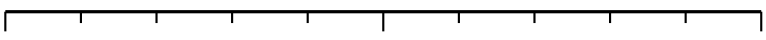
ROOMCLEANER FLOOR REFERENCE

sheet 1 of 9 - AprilTag 36h11 id 0

CENTRE SHEET: put it under the claw. It defines the floor frame:
+X = right as printed, +Y = up the page.







1. Print at ACTUAL SIZE / 100 % (never 'fit to page'). Print every sheet in one run, same settings.
2. With a ruler, MEASURE THE BLACK SQUARE EDGE of this sheet (printed: 150.0 mm). Enter the measured value of sheet 0, in metres, as marker.edge_m in the robot config. Check the 100 mm bar above on every sheet.
3. Reprint any sheet whose black square differs from sheet 0 by more than 1 mm.

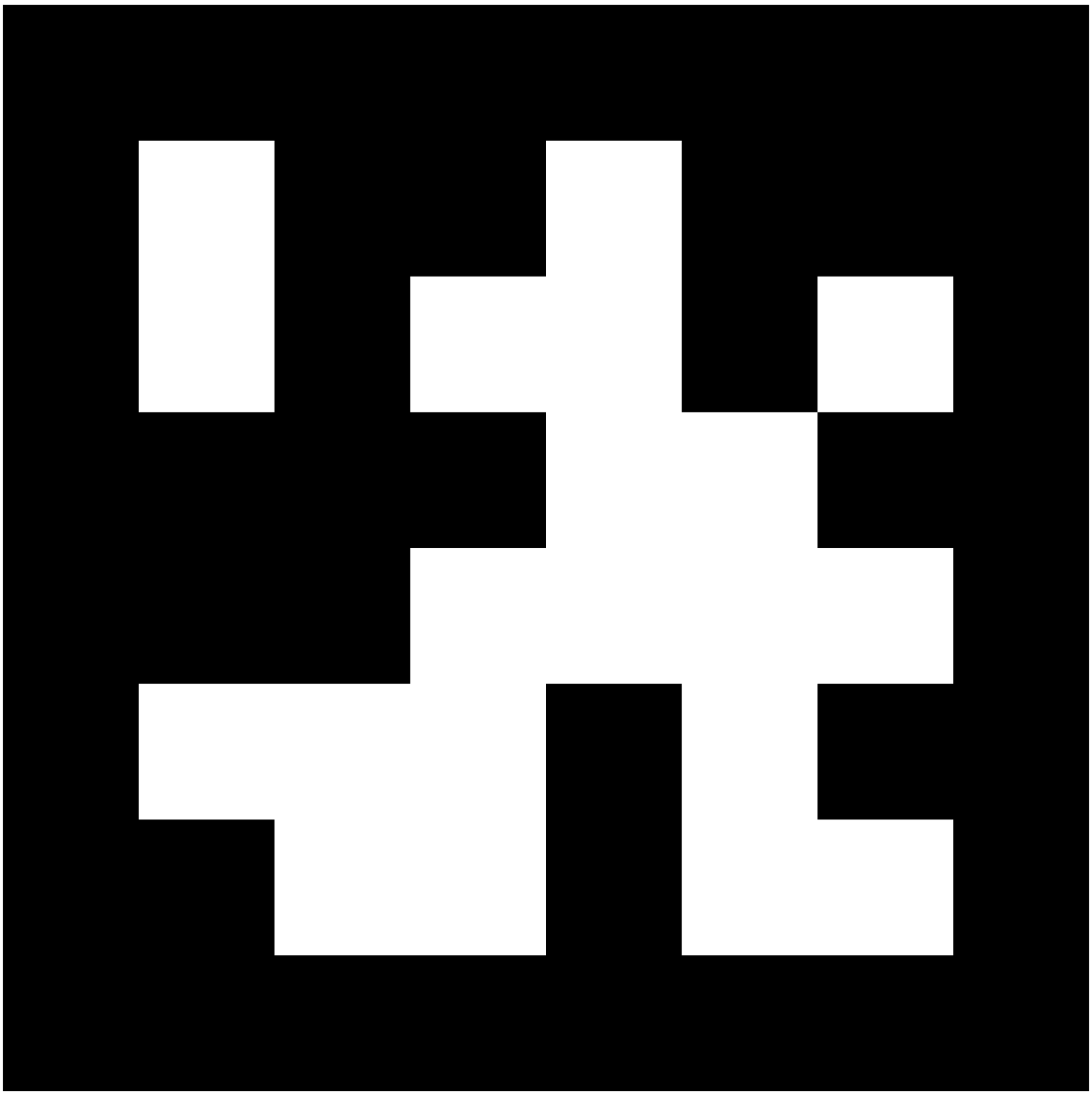
1

ROOMCLEANER FLOOR REFERENCE

sheet 2 of 9 - AprilTag 36h11 id 1

Lay flat, any rotation, roughly where the layout page shows.

TOP



<- must measure 100 mm

1. Print at ACTUAL SIZE / 100 % (never 'fit to page'). Print every sheet in one run, same settings.
2. With a ruler, MEASURE THE BLACK SQUARE EDGE of this sheet (printed: 150.0 mm). Enter the measured value of sheet 0, in metres, as marker.edge_m in the robot config. Check the 100 mm bar above on every sheet.
3. Reprint any sheet whose black square differs from sheet 0 by more than 1 mm.

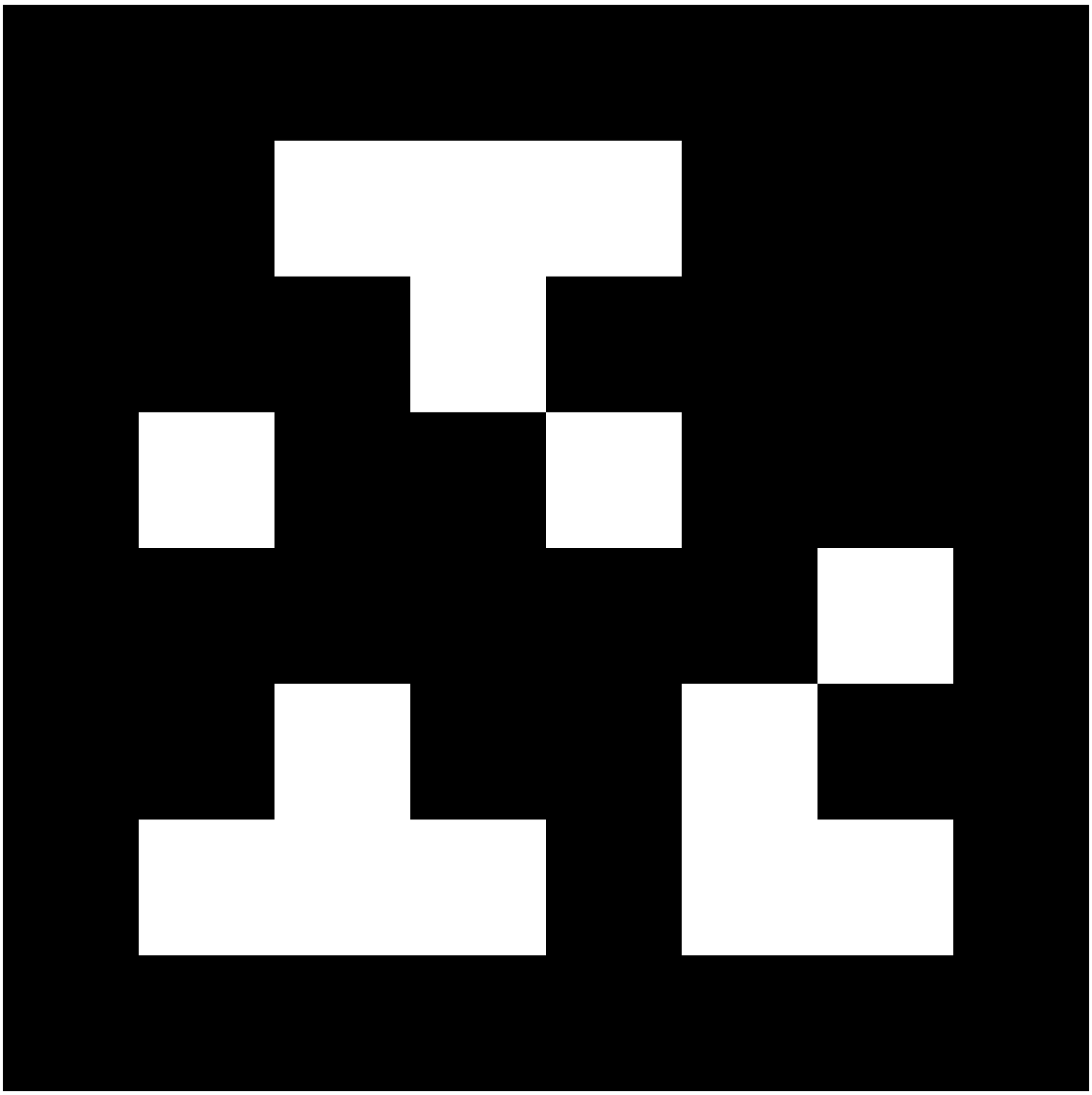
2

ROOMCLEANER FLOOR REFERENCE

sheet 3 of 9 - AprilTag 36h11 id 2

Lay flat, any rotation, roughly where the layout page shows.

TOP



<- must measure 100 mm

1. Print at ACTUAL SIZE / 100 % (never 'fit to page'). Print every sheet in one run, same settings.
2. With a ruler, MEASURE THE BLACK SQUARE EDGE of this sheet (printed: 150.0 mm). Enter the measured value of sheet 0, in metres, as marker.edge_m in the robot config. Check the 100 mm bar above on every sheet.
3. Reprint any sheet whose black square differs from sheet 0 by more than 1 mm.

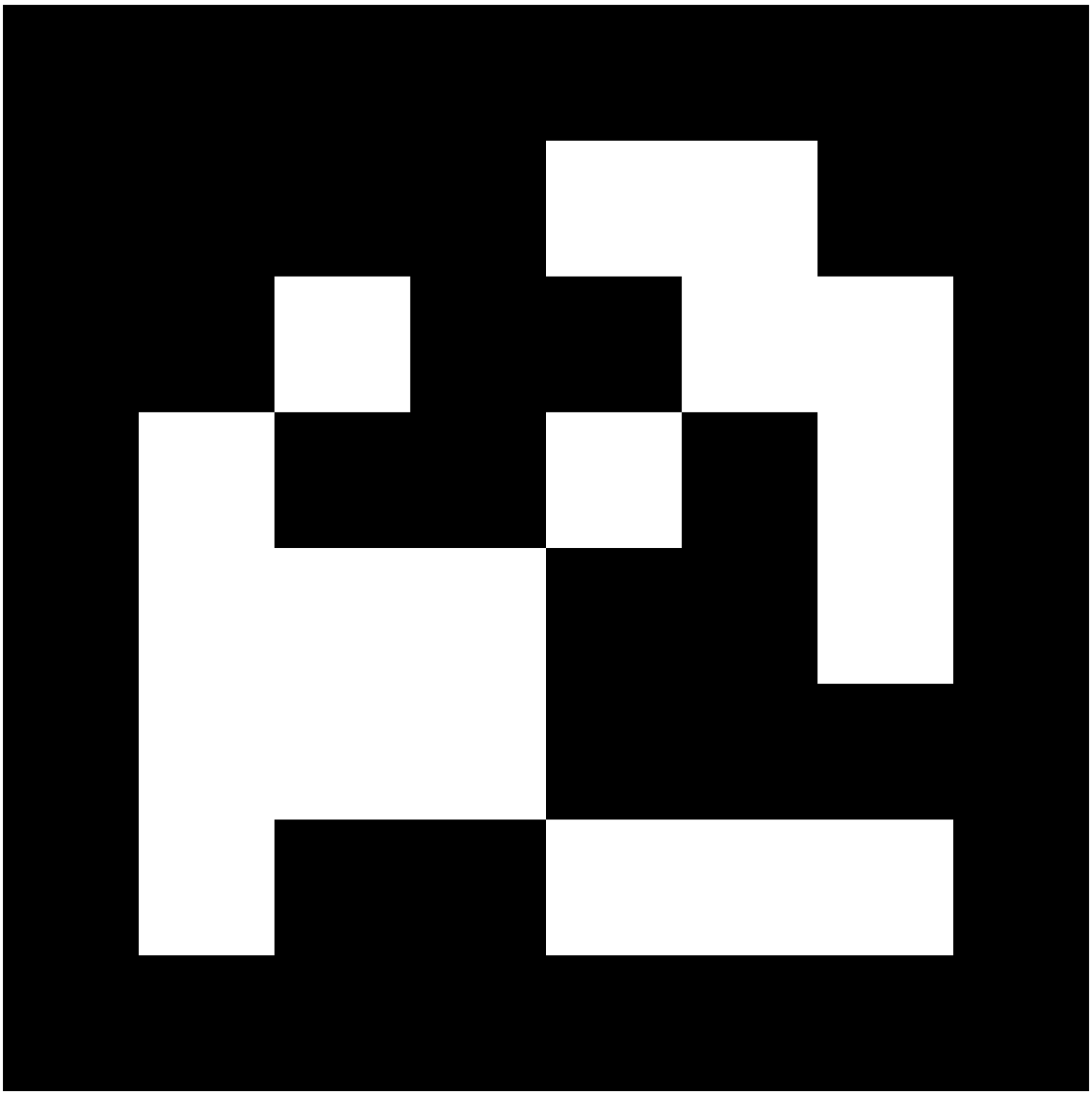
3

ROOMCLEANER FLOOR REFERENCE

sheet 4 of 9 - AprilTag 36h11 id 3

Lay flat, any rotation, roughly where the layout page shows.

TOP



<- must measure 100 mm

1. Print at ACTUAL SIZE / 100 % (never 'fit to page'). Print every sheet in one run, same settings.
2. With a ruler, MEASURE THE BLACK SQUARE EDGE of this sheet (printed: 150.0 mm). Enter the measured value of sheet 0, in metres, as marker.edge_m in the robot config. Check the 100 mm bar above on every sheet.
3. Reprint any sheet whose black square differs from sheet 0 by more than 1 mm.

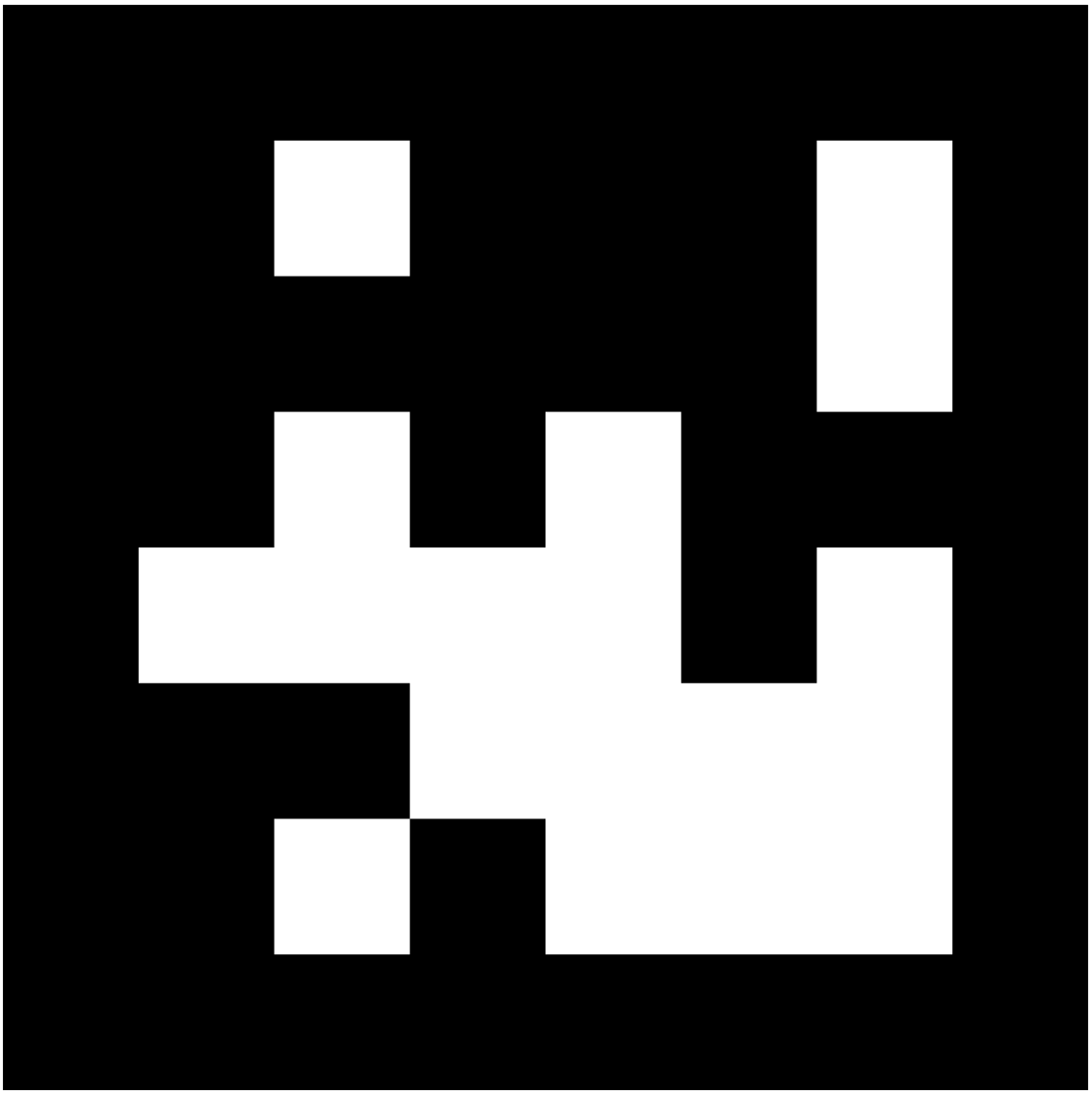
4

ROOMCLEANER FLOOR REFERENCE

sheet 5 of 9 - AprilTag 36h11 id 4

Lay flat, any rotation, roughly where the layout page shows.

TOP



<- must measure 100 mm

1. Print at ACTUAL SIZE / 100 % (never 'fit to page'). Print every sheet in one run, same settings.
2. With a ruler, MEASURE THE BLACK SQUARE EDGE of this sheet (printed: 150.0 mm). Enter the measured value of sheet 0, in metres, as marker.edge_m in the robot config. Check the 100 mm bar above on every sheet.
3. Reprint any sheet whose black square differs from sheet 0 by more than 1 mm.

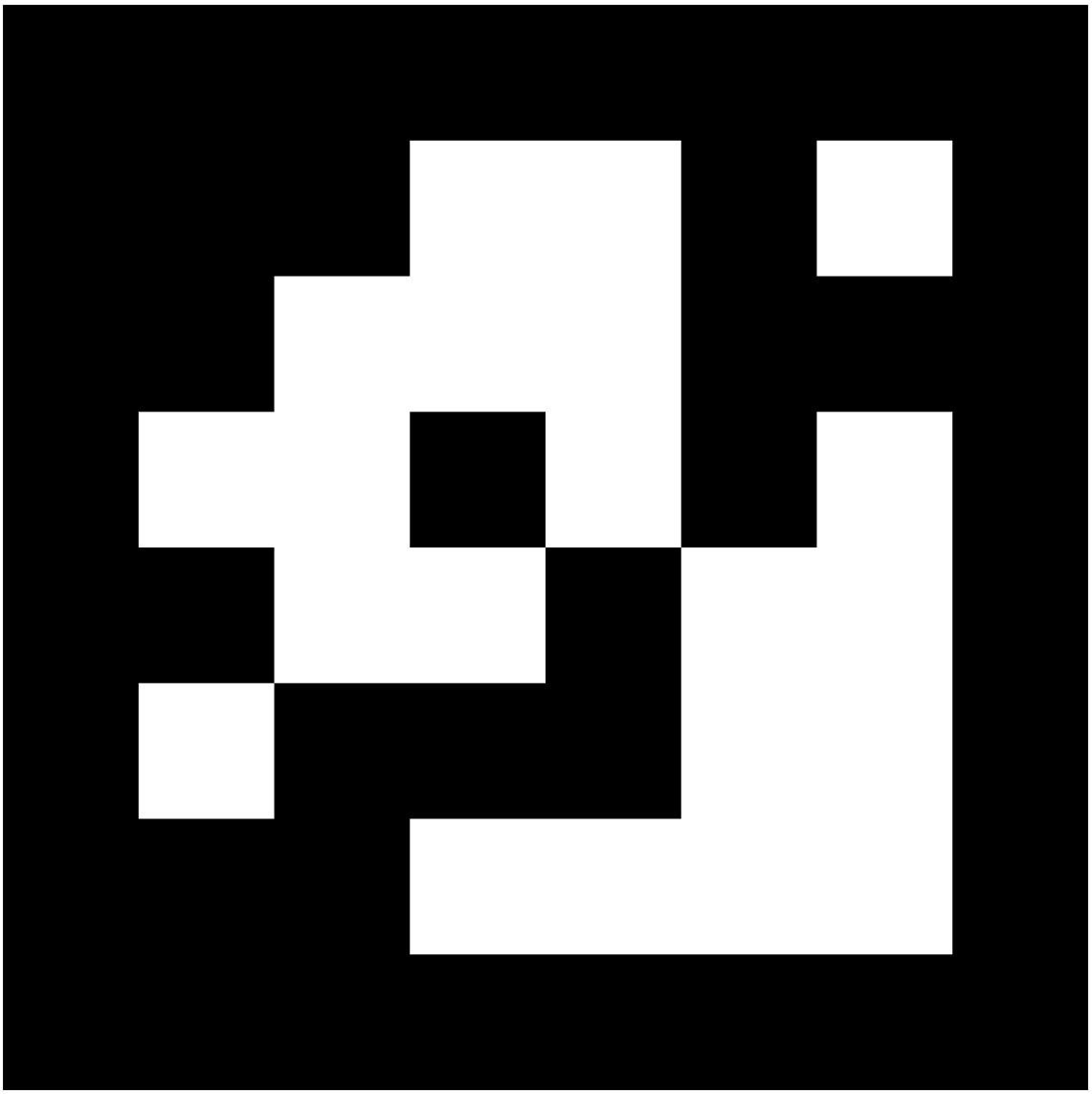
5

ROOMCLEANER FLOOR REFERENCE

sheet 6 of 9 - AprilTag 36h11 id 5

Lay flat, any rotation, roughly where the layout page shows.

TOP



<- must measure 100 mm

1. Print at ACTUAL SIZE / 100 % (never 'fit to page'). Print every sheet in one run, same settings.
2. With a ruler, MEASURE THE BLACK SQUARE EDGE of this sheet (printed: 150.0 mm). Enter the measured value of sheet 0, in metres, as marker.edge_m in the robot config. Check the 100 mm bar above on every sheet.
3. Reprint any sheet whose black square differs from sheet 0 by more than 1 mm.

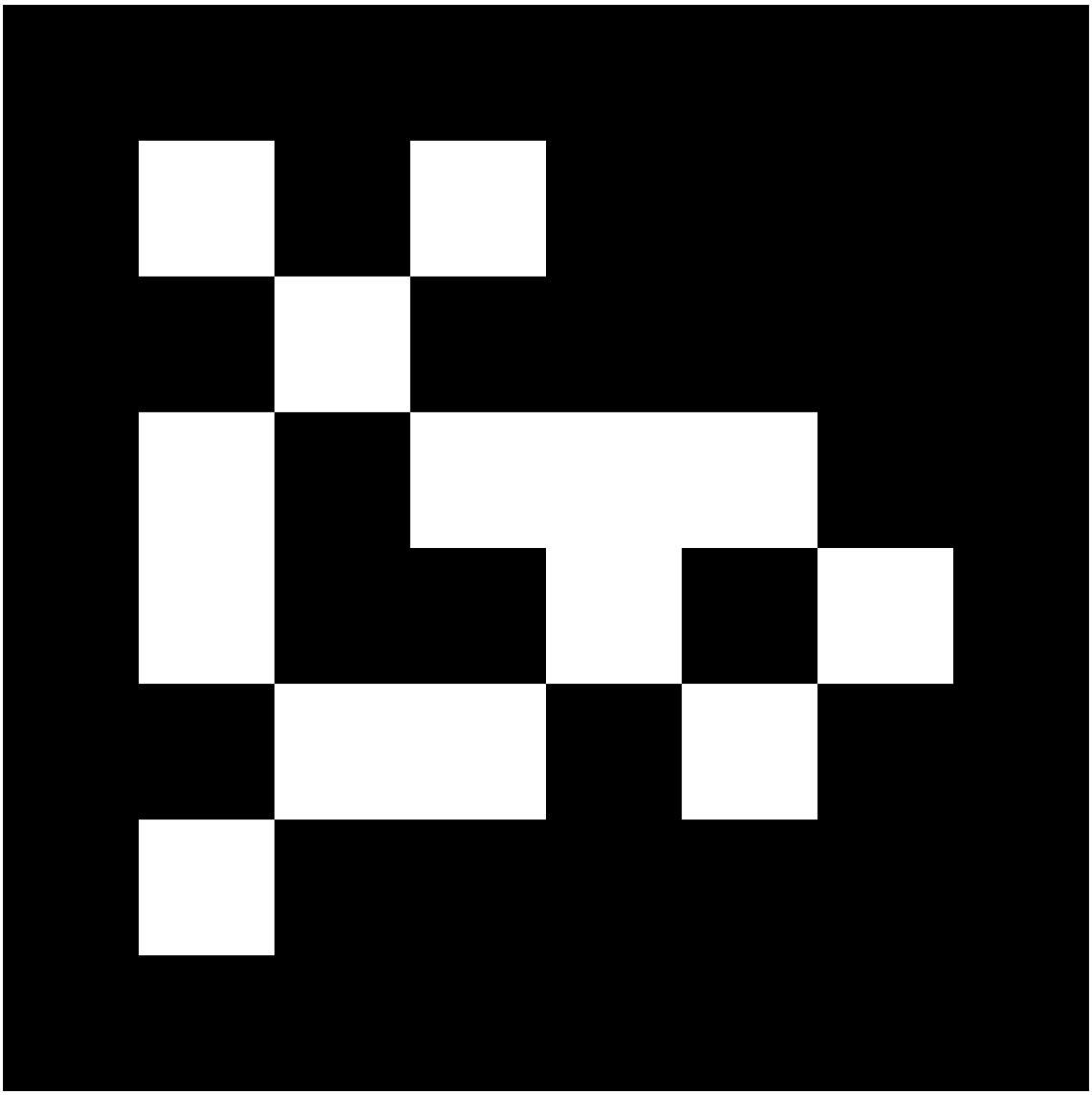
6

ROOMCLEANER FLOOR REFERENCE

sheet 7 of 9 - AprilTag 36h11 id 6

Lay flat, any rotation, roughly where the layout page shows.

TOP



<- must measure 100 mm

1. Print at ACTUAL SIZE / 100 % (never 'fit to page'). Print every sheet in one run, same settings.
2. With a ruler, MEASURE THE BLACK SQUARE EDGE of this sheet (printed: 150.0 mm). Enter the measured value of sheet 0, in metres, as marker.edge_m in the robot config. Check the 100 mm bar above on every sheet.
3. Reprint any sheet whose black square differs from sheet 0 by more than 1 mm.

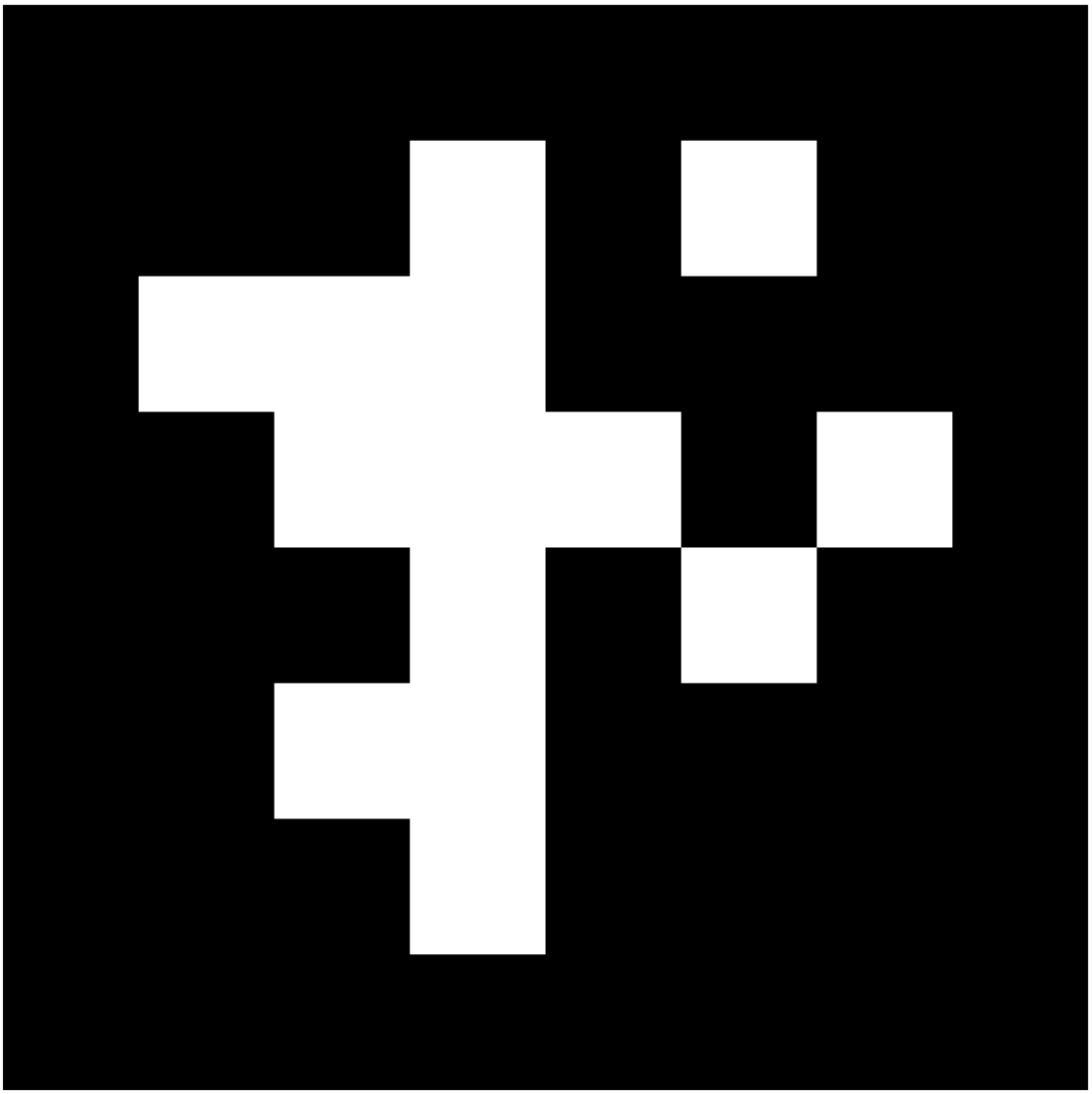
7

ROOMCLEANER FLOOR REFERENCE

sheet 8 of 9 - AprilTag 36h11 id 7

Lay flat, any rotation, roughly where the layout page shows.

TOP



<- must measure 100 mm

1. Print at ACTUAL SIZE / 100 % (never 'fit to page'). Print every sheet in one run, same settings.
2. With a ruler, MEASURE THE BLACK SQUARE EDGE of this sheet (printed: 150.0 mm). Enter the measured value of sheet 0, in metres, as marker.edge_m in the robot config. Check the 100 mm bar above on every sheet.
3. Reprint any sheet whose black square differs from sheet 0 by more than 1 mm.

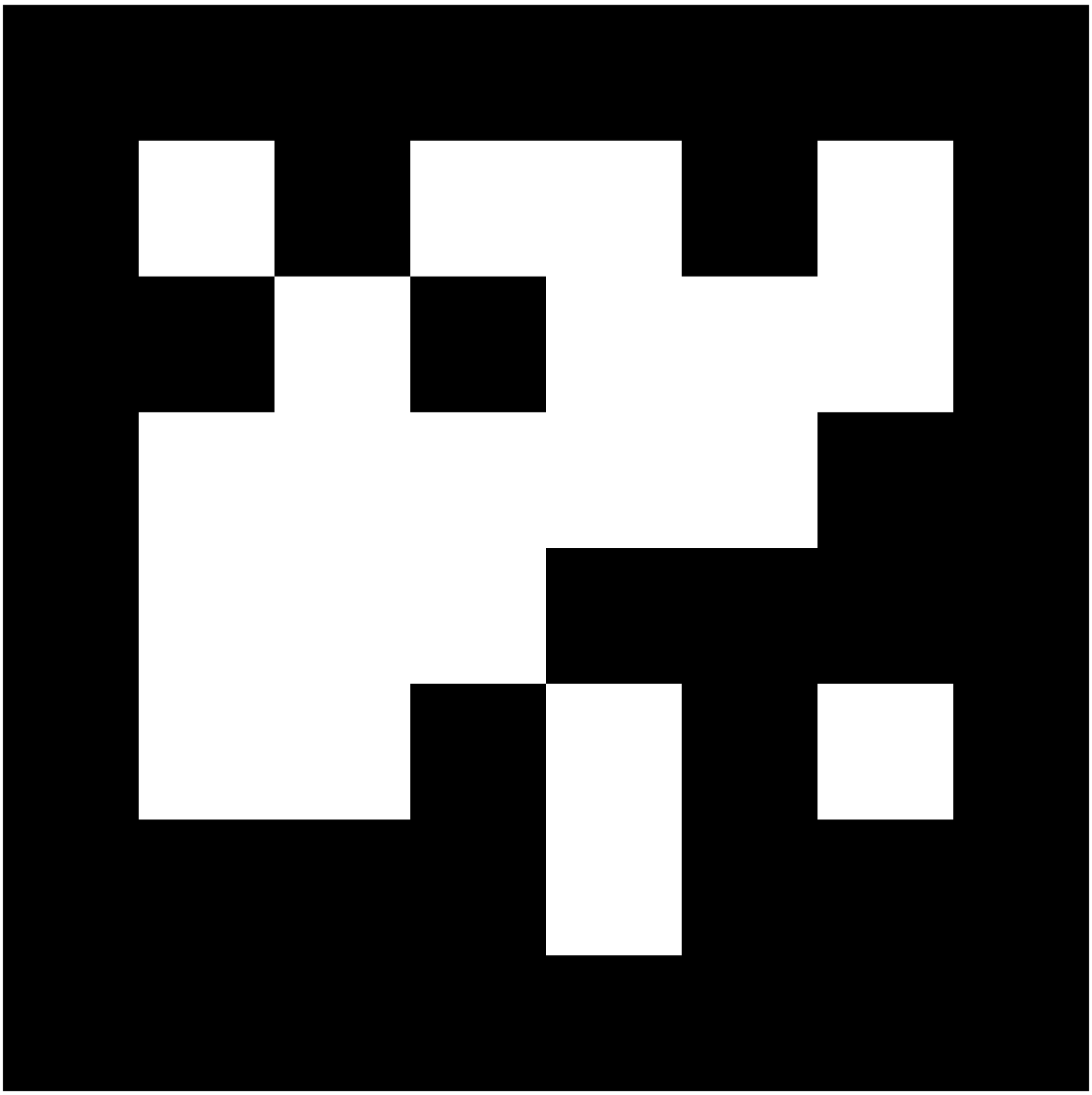
8

ROOMCLEANER FLOOR REFERENCE

sheet 9 of 9 - AprilTag 36h11 id 8

Lay flat, any rotation, roughly where the layout page shows.

TOP



<- must measure 100 mm

1. Print at ACTUAL SIZE / 100 % (never 'fit to page'). Print every sheet in one run, same settings.
2. With a ruler, MEASURE THE BLACK SQUARE EDGE of this sheet (printed: 150.0 mm). Enter the measured value of sheet 0, in metres, as marker.edge_m in the robot config. Check the 100 mm bar above on every sheet.
3. Reprint any sheet whose black square differs from sheet 0 by more than 1 mm.